

Homa Hosseinmardi

2328 Rolfe Hall
[University of California, Los Angeles](#)
[OASIS Lab](#)

homahm@ucla.edu
[Google Scholar](#)
+1 (720) 469-0696

Research topics Online and AI System's Safety, User Behavior, Socio-technical systems, Network Science, Statistical Inference

Positions Held

2024 - present	Assistant Professor, DataX and Communication Department Joint Appointment, University of California Los Angeles Director of Online and AI System's Integrity and Safety (OASIS Lab) Visiting scholar, Computational Social Science Lab, University of Pennsylvania
2020 - 2024	Senior Research Associate, Computational Social Science Lab, University of Pennsylvania
2023 - 2024	Visiting scholar, Digital Safety Research Institute
2019 - 2020	Postdoctoral Research Fellow, Computational Social Science Lab, University of Pennsylvania
2019	Machine Learning Scientist, Capital One
2017 – 2018	Postdoctoral Research Fellow, Information Sciences Institutes, University of Southern California
2015 - 2017	Research Scientist, Danaher Labs

Education

2012 - 2015	PhD, Computer Science, University of Colorado Boulder Thesis: <i>Multimodal Labeling and Characterization of Social Network Data for Detection and Prediction of Cyberbullying</i>
-------------	---

Selected Publications

- **H. Hosseinmardi**, S. Wolken, D. Rothschild, D. J. Watts, "The Growing Divide Between Cable and Network: Unpacking Media Bias on U.S. Television News." *Scientific Reports*, 2025.
- **H. Hosseinmardi**, A. Ghasemian, M. Rivera-Lanas, M. Ribeiro, B. West, D. J. Watts, "Causally Estimating the Effect of YouTube's Recommender System using Counterfactual Bots." *Proceedings of the National Academy of Sciences*, 2024.
- **H. Hosseinmardi**, A. Ghasemian, K. Lerman, and E. Ferrara, "Tensor Regression: A Supervised Framework for Human Behavioral Data Mining and Prediction." *IEEE ICHI* 2023.
- M. Ribeiro, **H. Hosseinmardi**, B. West, D. J. Watts, "Deplatforming did not Decrease Parler Users' Activity on Fringe Social Media." *PNAS nexus* 2, no. 3 (2023).
- D. Muise, **H. Hosseinmardi**, D. Rothschild, M. Mobius, D. J. Watts, "Quantifying Partisan News Diets in Web and TV Audiences." *Science advances* 8, no. 28 (2022).
- **H. Hosseinmardi**, A. Ghasemian, A. Clauset, D. Rothschild, Markus Mobius, D. J. Watts, "Examining the Consumption of Radical Content on YouTube." *Proceedings of the National Academy of Sciences* 118, no. 32, 2021.
- A. Ghasemian, **H. Hosseinmardi**, A. Galstyan, E. M. Airolidi, and A. Clauset. "Stacking Models for Nearly Optimal Link Prediction in Complex Networks." *Proceedings of the National Academy of Sciences* 117, no. 38, 2020.

Works in Progress

- ∞ A. Epps, M. Ekstrand, **H. Hosseinmardi**, N. Matias, “Navigating the Politics of Computing with Confidence and Purpose.” Work in progress.
- ∞ **H. Hosseinmardi**, U. Dutta, D. Rothschild, D. J. Watts, “Algorithmic Systems, Human Agency, and the Future of Platform Research.” Ready to submit.
- ∞ A. Ghasemian, **H. Hosseinmardi**, U. Dutta, D. Rothschild, D. J. Watts, “Supply and Demand of News-Like Information at the Rise of Individual Content Producers.” Ready to submit.
- ∞ P. Jamie, A. Ghasemian, **H. Hosseinmardi**, “Breaking the Bubble: Audit of Short Video Platforms.” Work in progress.
- ∞ A. Ghasemian, **H. Hosseinmardi**, U. Dutta, D. J. Watts, “Trapped by Personalization: LLM-Assisted Recommendations and Their Hidden Impact on Human-AI Ecosystems.” Ready to submit.
- ∞ P. Jamie, A. Razi, **H. Hosseinmardi**, “Auditing Rabbit Holes in Young Adolescents’ Online Sexual Information Seeking.” Work in progress.
- ∞ P. Savgira, **H. Hosseinmardi**, “What Stays and What Goes? Evaluating LLM Summaries and Domain Specific Information Reduction.” Work in progress.
- ∞ X. Xi, S. Rosskopf, O. Nguyen, A. Ghasemian, **H. Hosseinmardi**, “News-like Ecosystem Benchmark: A YouTube case study.” Work in progress.
- ∞

Grants

- Co-PI, \$1.5 M , Huo Family Foundation (HFF), 2026–2029, “Beyond Screen Time: Investigating the Influence of Social Media Content and Algorithms on Adolescent Mood Using Multimodal Longitudinal Data and Counterfactual Bots.”
- New faculty awards, UCLA, \$5,000, 2025-2026.
- Co-PI, \$20,000 Google Award, 2023-2024, “Modeling YouTube Users’ Production and Consumption of News.”
- Allocation Manager, \$100,000 - COVID19 HPC Consortium award, “COVID Mapping and Modeling for City of Philadelphia.”

Work Under Review

- ◇ U. Dutta, **H. Hosseinmardi**, A. Ghasemian, A. Clauset, D. J. Watts, “Asymmetric Framing Strategies Drive Partisanship in News.” Under Review.
- ◇ “Mapping and Auditing Cancer Information Ecosystems on Social Media at Scale to Understand User Engagement and Platform Dynamics,” A. Ghasemian, **H. Hosseinmardi**, S. Rezapour, P. Ganz, Under Review.
- ◇ A. Ghasemian, **H. Hosseinmardi**, S. Rezapour, P. Jamie “The Effectiveness of LLM Agents Beyond Individual Belief Change: Toward Group and Networked Collective Dynamics,” under review.
- ◇ **H. Hosseinmardi**, A. Ghasemian, P. Jamie “Coordinated Multi-Turn Adversarial Red Teaming for Multi-Agent Systems,” under review.

Professional Services

- ♠ Editor: Journal of Quantitative Description: Digital Media
- ♠ Journal review: Nature, Science, PNAS, Science Advances, EPJ Data Science, PLOS ONE, and Future Generation, Computer Systems Journal, Computers in Human Behavior Journal.
- ♠ Program Chair, The International Conference for Computational Social Science, 2024.
- ♠ Co-founder of CU CyberSafety Research Center and workshop series on Computational Methods for CyberSafety.
- ♠ Organizing the First International Workshop on Computational Methods for CyberSafety, CIKM 2016. a Technical Program Committee Co-chair, International Workshop on Computational Methods for CyberSafety, WWW 2017, 2018, 2020.
- ♠ Workshop Co-chair, The Fourth Workshop on Computational Methods in Online Misbehavior, WWW 2019.
- ♠ Program committee: The Web Conf (WWW), WSDM, NetSci, IC2S2, ASONAM, WiDS, DSAA 2024
- ♠ Conference reviewer: ICWSM, ICDM, SDM

Recent Talks

- δ Invited speaker, Panelist, ICA (June 2026)
- δ Keynote speaker, [Computational Journalism Symposium](#), University of Miami (Dec 2025)
- δ Invited speaker, [Data Justice talk series](#), UCLA (Dec 2025)
- δ Invited speaker, the National Student Data Corps, UCLA (Nov 2025)
- δ Invited speaker, [ISI, USC](#) (Nov 2025)
- δ Invited speaker, [Summer School, Computational Social Science](#), UCLA (Aug 2025)
- δ Invited speaker, Meta research, Computational Social Science team, DC (July 2025)
- δ Panelist, the National Student Data Corps, UCLA (May 2025)
- δ Panelist, DataX+IDRE AI Symposium, UCLA (April 2025)
- δ Invited speaker, NYU Tech & Society Symposium (April 2025)
- δ Invited speaker, USC (March 2025)
- δ Invited speaker, American Museum of Natural History (Dec 2024)
- δ Invited speaker, National Academy Science workshop series (Sep 2024)
- δ Panelist, Political and Information Networks Workshop (April 2024)
- δ Seminar series, University of Virginia (March 2024)
- δ Seminar series, University of Arizona (March 2024)
- δ Seminar series, University of Binghamton (March 2024)
- δ Seminar series, University of Louisiana (March 2024)
- δ Seminar series, Drexel University (Feb 2024)
- δ Seminar series, NYU's Center for Social Media and Politics (Dec 2023)
- δ Invited speaker, Conference on Harmful Online Communication (CHOC 2023), GESIS, Germany (Nov 2023)
- δ Invited speaker, Fairness and Safety Speaker Series, YouTube, NYC (Sep 2023)
- δ Invited speaker, the Wharton School's Women in Data Science Conference (Feb 2023)
- δ Seminar series, University of California, Los Angeles (Jan 2023)
- δ Invited Moderator, NYU's Center for Social Media and Politics Seminar Series, "Media Consumption, Misinformation, and Polarization" (Oct 12, 2022)
- δ Invited speaker, Statistical Inference for Network Models satellite (July 2022)
- δ Seminar series, Boston University (March 2022)
- δ Seminar series, Georgetown University (Feb 2022)
- δ Seminar series, Cornell University (Feb 2022)
- δ Seminar series, Georgia Institute of Technology (Jan 2022)
- δ Seminar series, University of Illinois-Urbana Champion (Jan 2022)
- δ Beyond Misinformation Workshop (Aug 2021)

Media Coverage

- ∞ ["How cable news has diverged from broadcast news,"](#) (June 2025) Penn Today.
- ∞ "To Create Boundaries With Social Media, We Must Understand How It Affects Us," Interviewed by Kirsten Brehmer, [Daily Bruin](#) (March 9, 2025).
- ∞ The YouTube Algorithm Isn't Radicalizing People: Why User Choice Matters on Social Media, [Knowledge at Wharton](#) (April 2024).
- ∞ [Estimating the Effect of YouTube Recommendations with Homa Hosseinmardi](#), Research Spotlight — Featured in Wharton Women in Analytics (March 2024).
- ∞ "Does YouTube's Algorithm Radicalize Young Americans? New Study Has Answers" — Research on the causal role of recommendation algorithm in [TECH TIMES](#) (February 21, 2024).
- ∞ ["The YouTube algorithm isn't radicalizing people,"](#) (February 2024) Penn Today.

- ∞ “People, not Google’s algorithm, create their own partisan ‘bubbles’ online” (May, 2023) — Featured in [Penn Today](#).
- ∞ “People, Not Google’s Algorithm, Create Their Own Partisan ‘Bubbles’ Online” [Scientific American](#) (2023).
- ∞ “Cable News Confusion” (September 27, 2022) — Research presented in talk at PolitiFact’s virtual [United Facts of America conference](#).
- ∞ “Comment: It’s not social media that’s polarizes; it’s cable news,” [Herland Net](#) (August 21, 2022).
- ∞ “It’s not just social media: Cable news has bigger effect on polarization when it comes to echo chambers, TV trumps online,” [Ars Technica](#) (August 10, 2022).
- ∞ “YouTube pushed Trump supporters toward voter fraud videos, study finds” (September 1, 2022) — Featured as an expert counterpoint opinion in [NBC News](#).
- ∞ “TV Echo Chambers and Political Polarization” (August 18, 2022) — Broadcast live on Sydney’s 2SER 107.3FM, research was the central topic of the [Fourth Estate](#) radio show
- ∞ “Homa Hosseinmardi on TV, Social Media and Political Echo Chambers” (August 17, 2022) — Featured in [Penn Engineering Today](#).
- ∞ “Don’t be too quick to blame social media for America’s polarization – cable news has a bigger effect, study finds” (August 10, 2022) — Original research op-ed featured in [The Conversation US](#)
- ∞ “TV news top driver of political echo chambers in U.S.” (July 26, 2022) — Featured in [Penn Today](#)
- ∞ “A Platform ‘Weaponized’: How YouTube Spreads Harmful Content—And What Can Be Done About It” (June 2022) — Research cited in [NYU Stern Center for Business and Human Rights’ report](#) on misinformation on YouTube
- ∞ “Wharton Women in Data Science and Analytics” (March 6, 2022) — Featured in the Wharton School’s list of [Women Creating Social Impact with Analytics](#)
- ∞ “Misinformation: What Should Our Tech Overlords Do?” (February 24, 2022) — Interviewed by Gimlet Media’s [Science VS](#) podcast
- ∞ “What Big Data Reveals About Online Extremism” (November 2021) — Interview/spotlight piece featured in [Annenberg School for Communication news](#)
- ∞ Research cited in congress (October 2021) — “Evaluating the scale, growth, and origins of right-wing echo chambers on YouTube” cited in an expert testimony in a [hearing of the U.S. Senate Committee on Homeland Security and Governmental Affairs](#)
- ∞ “If YouTube’s algorithms radicalize people, it’s hard to tell from the data” (August 2021) — Research on the consumption of radical content on YouTube covered in [Ars Technica](#)
- ∞ “YouTube Chief Product Officer Neal Mohan on the Algorithm, Monetization, and the Future for Creators” (August 2021) — Research on the consumption of radical content on YouTube covered in [The Verge](#)
- ∞ “Studie: Youtube radikalisiert User nicht systematisch” (August 3, 2022) — Interviewed by German radio and journal [Deutschlandfunk](#)

Teaching and Mentorship

Guest lectures

- ✓ Computational Ethics, Large Language Models and the Future of NLP, Computer Science, UCLA (grad, Winter 2025)
- ✓ LLM for Social Good, Communication Department, UCLA (undergrad, Fall 2024)
- ✓ Ethics and Activism in Tech and Design, Computer Science, UCSC (grad, Spring 2024, Winter 2025)
- ✓ Probabilistic Models of Human and Machine Intelligence, Computer Science, UC-Boulder (grad, Fall 2015)
- ✓ Fundamentals of Human Computer Interaction, Computer Science, UC-Boulder (undergrad, Spring 2014)

Teaching

- ✓ Instructor, Data Science for Good (ungrad, Spring 2026)
- ✓ Instructor, Applied Machine Learning (grad, Spring 2026)
- ✓ Instructor, Cambridge Center for International Research (Summer 2025)
- ✓ Instructor, Data Science for Social Good (ungrad, Spring 2025)
- ✓ Instructor, Algorithms and Society (grad, Spring 2025)
- ✓ Instructor, Cambridge Center for International Research (Summer-Fall 2022)

- ✓ Danaher Labs, Data Science bootcamps (Spring 2016, Fall 2017)
- ✓ Teaching Assistant for undergraduate signal processing course, University of Tehran, Fall 2009, Spring 2010
- Mentorship*
 - ♣ (PhDs) Pooriya Jamie, Pavel Savgira, Upasana Dutta, Bethany Hsiao, Chloe Ahn, Sam Wolken, Hsien-Te Kao, Shen Yan, Rahat Ibn Rafigh;
 - ♣ (MSs) Yue Chen, Rachel S. M. Leong, Zhangyi Fan, Xiaonan Liu, Keith Golden, Kailun Li, Tai Nguyen, Yihong Zhang, Robby Qiu;
 - ♣ (UGs) Sean Roskopf, Olivia Nguyen, Xiaotong (Riley) Xi, Valérie Ding, Ovin Huang, Vidusha Adira, Josie Arbuckle, Josh Ludan, Misty Liao, Seven Fandozzi, Vivienne Chen, Henry Ge, Anna Callahan;
 - ♣ (Google Summer of Code Intern) Karanjot Singh;
 - ♣ (Danaher Corporation) Shaghayegh Kazemlu, Abbas Hooshmand, Peng Wang, Lisa Norgard, Eliyar Asgharieh.
 - ♣ Summer School on Methods for Computational Social Science, ISI, USC, Summer 2018

Awards

- University of Colorado Boulder, Department of Computer Science Outstanding Research Award, May 2015.
- University of Colorado Boulder, Department of Computer Science Graduate Student Fellowship, Fall 2012.

Other Experiences

- [YouTube Politics, Living Journal.](#)
- [Measuring the Information Ecosystem](#)
- Analysis of the mobility data of Covid-19 for the City of Philadelphia: Optimal Interventions to Control COVID-19 in Networked Populations, J. F. Barreras, V. Preciado, H. Hassani, D. J. Watts, **H. Hosseinmardi**, A. Ghasemian, M. Whiting.

More Publications

- N. Tavabi, **H. Hosseinmardi**, J. L. Villatte, A. Abeliuk, S. Narayanan, E. Ferrara, and K. Lerman. “Learning Behavioral Representations from Wearable Sensors.” In International Conference on Social Computing, Behavioral-Cultural Modeling and Prediction and Behavior Representation in Modeling and Simulation, pp. 245-254. Springer, Cham, 2020.
- H. Kao, S. Yan, **H. Hosseinmardi**, S. Narayanan, K. Lerman, and Emilio Ferrara. “User-Based Collaborative Filtering Mobile Health System.” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 4, no. 4, 1-17, 2020.
- S. Yan, **H. Hosseinmardi**, H. T. Kao, S. Narayanan, K. Lerman, E. Ferrara, “Affect Estimation with Wearable Sensors.” Journal of Healthcare Informatics Research, 1-34, 2020.
- A. Ghasemian, **H. Hosseinmardi**, and A. Clauset, “Evaluating Overfit and Underfit in Models of Network Community Structure.” *IEEE Transactions on Knowledge and Data Engineering* 32, no. 9 (2019): 1722-1735.
- H. T. Kao, S. Yan, D. Huang, N. Bartley, **H. Hosseinmardi**, E. Ferrara, “Understanding Cyberbullying on Instagram and askfm via Social Role Detection.” In Companion Proceedings of the 2019 World Wide Web Conference, pp. 183-188. 2019.
- S. Yan, **H. Hosseinmardi**, H. T. Kao, S. Narayanan, K. Lerman, E. Ferrara, “Estimating individualized daily self-reported affect with wearable sensors.” In 2019 IEEE International Conference on Healthcare Informatics (ICHI), pp. 1-9. IEEE, 2019.
- H. T. Kao, **H. Hosseinmardi**, S. Yan, M. Hasan, S. Narayanan, K. Lerman and E. Ferrara, “Discovering Latent Psychological Structures from Self-report Assessments of Hospital Workers.” International Conference on Behavioral, Economic, and SocioCultural Computing (BESC2018) (best paper award).
- P. Goyal, H. Hosseinmardi, E. Ferrara, A. Galstyan, “Capturing edge attributes via network embedding.” *IEEE Transactions on Computational Social Systems* 5, no. 4 (2018): 907-917.
- **H. Hosseinmardi**, E. Ferrara, K. Lerman, “Discovering Hidden Structure in High Dimensional Human Behavioral Data via Tensor Factorization.” *Hetronam, WSDM* 2018.
- P. Goyal, **H. Hosseinmardi**, E. Ferrara, A. Galstyan, “Embedding networks with edge attributes.” In Proceedings of the 29th on Hypertext and Social Media, pp. 38-42, 2018.

- R. Ibn Rafiq, **H. Hosseinmardi**, R. Han, Q. Lv, S. Mishra, "Scalable and timely detection of cyberbullying in online social networks." In Proceedings of the 33rd Annual ACM Symposium on Applied Computing, pp. 1738-1747. 2018.
- **H. Hosseinmardi**, R. Ibn Rafiq, R. Han, S. Mishra, Q. Lv, "Prediction of Cyberbullying Incidents in a Media-based Social Network." In 2016 *IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining* (ASONAM), pp. 186-192. IEEE, 2016.
- R. Ibn Rafiq, **H. Hosseinmardi**, S. A. Mattson, R. Han, Q. Lv, S. Mishra, "Analysis and detection of labeled cyberbullying instances in Vine, a video-based social network." *Social network analysis and mining* 6, no. 1, 1-16, 2016.
- **H. Hosseinmardi**, R. Ibn Rafiq, S. A. Mattson, R. Han, S. Mishra, Q. Lv, "Analyzing Factors Impacting Reviving on the Vine Social Network." In *International Conference on Social Informatics*, pp. 17-32. Springer, Cham, 2015.
- R. Ibn Rafiq, **H. Hosseinmardi**, R. Han, Q. Lv, S. Mishra, and S. A. Mattson. "Careful what you share in six seconds: Detecting cyberbullying instances in Vine." In 2015 *IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining* (ASONAM), pp. 617-622. IEEE, 2015.
- **H. Hosseinmardi**, R. Ibn Rafiq, R. Han, S. Mishra, Q. Lv, "Analyzing Labeled Cyberbullying Incidents on the Instagram Social Network." In *International conference on social informatics*, pp. 49-66. Springer, Cham, 2015.
- **H. Hosseinmardi**, N. Correll, and R. Han, "Distributed Spatio-Temporal Gesture Recognition in Sensor Arrays." *ACM Transactions on Autonomous and Adaptive Systems* (TAAS) 10, no. 3, 1-19, 2015.
- **H. Hosseinmardi**, A. Ghasemian, R. Han, Q. Lv, and S. Mishra, "Towards Understanding Cyberbullying Behavior in a Semi-anonymous Social Network." In 2014 *IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining* (ASONAM 2014), pp. 244-252. IEEE, 2014.
- **H. Hosseinmardi**, R. Ibn Rafiq, S. Li, Z. Yang, R. Han, Q. Lv and S. Mishra, "A Comparison of Common Users across Instagram and ask.fm to Better Understand Cyberbullying." In 2014 *IEEE Fourth International Conference on Big Data and Cloud Computing*, pp. 355-362. IEEE, 2014.
- S. Ma, **H. Hosseinmardi**, N. Farrow, R. Han, and N. Correll, "Establishing Multicast Groups in Computational Robotic Materials." In 2012 *IEEE International Conference on Green Computing and Communications*, pp. 311-316. IEEE, 2012.
- **H. Hosseinmardi**, R. Han, and N. Correll, "Bloom Filter-Based Ad Hoc Multicast Communication in Cyber-Physical Systems and Computational Materials." In *International Conference on Wireless Algorithms, Systems, and Applications*, pp. 595-606. Springer, Berlin, Heidelberg, 2012.
- **H. Hosseinmardi**, and F. Lahouti, "Multicast lifetime maximization using network coding: A cross-layer approach." In 2008 *24th Biennial Symposium on Communications*, pp. 1-4. IEEE, 2008.
- **H. Hosseinmardi**, and F. Lahouti, "Online multicasting using network coding in energy constrained wireless ad hoc networks." In 2008 *3rd International Symposium on Wireless Pervasive Computing*, pp. 545-549. IEEE, 2008.